

# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India**

**09/2021 – 01/2025 (3.5 Years)**

- Developed and validated **R-Shiny statistical analysis programs** for clinical trial data interpretation, collaborating with biostatisticians to translate study requirements into accurate, reproducible analytical implementations with **100% compliance** to ICH/GCP guidelines.
- Prepared and reviewed **statistical analysis plans (SAPs), programming specifications, and derived datasets** across enterprise clinical systems, ensuring data integrity and regulatory standards adherence with **zero defects** in UAT cycles.
- Executed end-to-end statistical programming workflows in **R and Python** (pandas, NumPy, scikit-learn) for clinical data management systems, performing exploratory data analysis, hypothesis testing, and time-series modeling to support clinical decision-making across **15+ trial submissions**.
- Mentored junior statistical programmers and cross-functional teams on **R programming techniques, best practices, and statistical methodologies**, delivering training materials and code-review standards that reduced peer-review cycles by **30%** and improved code quality metrics.
- Ensured **data integrity and compliance with industry regulations including ICH/GCP guidelines and GDPR** throughout statistical programming processes, implementing change-control workflows and audit-trail documentation across clinical trial datasets supporting regulatory submissions.
- Collaborated with biostatisticians, clinical operations teams, and regulatory stakeholders to interpret complex study requirements, validate statistical outputs, and communicate analytical findings to non-technical stakeholders, achieving **98% first-pass accuracy** in SAP implementations.

## SKILLS

**Statistical Programming & Clinical Analysis** – R, R-Shiny, Python (pandas, NumPy, scikit-learn), statistical methodologies, exploratory data analysis, hypothesis testing, time-series analysis, machine learning fundamentals.

**Clinical Data Management & Regulatory Compliance** – Statistical Analysis Plans (SAPs), programming specifications, derived datasets, data integrity validation, ICH/GCP guidelines, GDPR, clinical trial data management systems, regulatory standards.

**Data Visualization & Business Intelligence** – R-Shiny dashboards, Power BI, Tableau, interactive dashboards, DAX measures, data modelling, Power Query, statistical reporting.

**Programming & Software Development** – Python, R, SQL (PostgreSQL, MySQL, Oracle), REST APIs, Git, CI/CD pipelines, code review, agile methodologies.

**Data Engineering & Database Management** – SQL, PostgreSQL, MySQL, Oracle, data warehousing, ETL workflows, data pipelines, schema design, ad-hoc query optimization.

**Collaboration & Communication** – Cross-functional team coordination, stakeholder engagement, technical documentation, training delivery, mentoring, complex concept communication to non-technical audiences.

**Infrastructure & Tools** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Jira, Confluence, Splunk, Dynatrace.

## EDUCATION AND TRAINING

## PROJECTS

### **Clinical Trial Statistical Analysis & R-Shiny Reporting Platform (HCL Technologies)**

- Architected and developed an **end-to-end R-Shiny statistical analysis platform** supporting **12 concurrent Phase II/III clinical trials**, integrating **SQL-based derived datasets, Python data pipelines, and R statistical models** to automate SAP execution and regulatory reporting.
- Implemented **interactive statistical dashboards in R-Shiny** that enabled biostatisticians and clinical teams to explore trial outcomes in real-time, reducing manual analysis time by **45%** and improving decision-velocity for protocol amendments by **2 weeks**.
- Designed and validated **statistical programming specifications and data lineage documentation** ensuring **100% ICH/GCP and GDPR compliance** across all clinical datasets, with automated audit-trail logging and change-control workflows supporting regulatory inspection readiness.
- Mentored **3 junior statistical programmers** on R best practices, exploratory data analysis techniques, and hypothesis-testing methodologies, establishing code-review standards and training curriculum that improved peer-review efficiency by **30%**.
- Executed complex **time-series analysis, survival modeling, and multivariate statistical testing** using **Python scikit-learn and R base libraries**, delivering **18 regulatory submissions** with **98% first-pass accuracy** across safety and efficacy endpoints.

### **Clinical Data Integrity & Compliance Automation Framework (HCL Technologies)**

- Developed **Python ETL pipelines and SQL data validation scripts** to enforce **ICH/GCP and GDPR compliance** standards across clinical trial datasets, automating **200+ validation rules** and eliminating manual compliance audits by **60%**.
- Built **statistical reporting dashboards in Power BI and Tableau** to visualize data quality metrics, regulatory compliance status, and derived dataset lineage for clinical operations and audit teams, enabling real-time visibility into **8 active trials**.
- Collaborated with biostatisticians and regulatory stakeholders to document **statistical analysis plans and programming specifications**, translating clinical requirements into validated R and Python analytical workflows with **zero defect rates** in UAT cycles.

## ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)